

Belief or Action? Semantic Ambiguity in the Italian Non-finite Domain

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Abstract: This paper examines the belief/intention semantic alternation found in Italian clause-embedding verbs like *convincere* ('convince'), a phenomenon that is typically attributed to finiteness or mood (Giannakidou & Mari 2021, Grano 2018, 2024). Interestingly, Italian exhibits this alternation within its non-finite domain: *di*-infinitives convey belief readings, while *a*-infinitives give rise to intention readings. We argue that this semantic distinction corresponds directly to the structural size of the complement clauses. Syntactic evidence shows that *di*-infinitives project larger structures, accommodating modal auxiliaries and allowing both subject and object control, consistent with a propositional interpretation and the presence of a CLOSURE operator. In contrast, *a*-infinitives are structurally smaller, lack higher functional projections, and allow only object control. Our semantic analysis proposes that *a*-infinitives select an event-denoting complement whose eventuality argument remains unsaturated, thereby establishing a direct causal link to the matrix attitude, in line with intention-encoding accounts (Grano 2024), whereas *di*-infinitives, selecting larger structures, are fully propositional.

Keywords: *rational attitudes, belief/intention alternation, infinitival complements*

1 Introduction

Crosslinguistically, it has been observed that certain clause embedding verbs exhibit a belief/intention semantic alternation with respect to their complement clause (Giannakidou & Mari 2021, Grano 2018, 2024). Such alternation correlates with specific morpho-syntactic features of the complement clause, for example finiteness for English *persuade* (Grano 2019, 2024) and Russian *dumat* (1), indicative vs. subjunctive mood for Greek *pitho* 'persuade' (2).

- (1) a. Vasja *dumaet* *vypit'* *piva*.
 Vasja think-PRS.3SG drink-INF beer.GEN
 'Vasja intends to drink beer.'
- b. Vasja *dumaet* *čto* *my idem* *pit'* *pivo*.
 Vasja think-PRS.3SG that we go-PRS.1PL drink-INF beer.ACC
 'Vasja thinks that we are going to drink beer.'
- (Russian, Kasenov 2023)
- (2) a. I *Ariadne epise* *ton Nikolas na* *fiji* *noris*.
 the Ariadne persuade-PST.3SG the Nikolas SBJV leave-PRS.3SG early
 'Ariadne persuaded/convinced Nikolas to leave early.'
- b. I *Ariadne epise* *ton Nikolas oti* *i* *idea tou ine* *kali*.
 the Ariadne persuade-PST.3SG the Nikolas that.IND the idea his be.PRS.3SG good
 'Ariadne persuaded/convinced Nikolas that his idea is good.'
- (Modern Greek, Giannakidou & Mari 2021)

Interestingly, Italian *convincere* 'convince' (alongside other verbs, like *pensare* 'think', and *avere paura* 'fear'), shows the same alternation within the non-finite domain. The verb can select two infinitival prepositional complementizers:

- (3) a. Marco *ha* *convinto* *Gianni di* *avere* *un figlio*.
 Marco₁ have-PRS.3SG convince-PST.PTCP Gianni₂ *di* PRO_{1/2} have-INF a child
 'Marco has convinced Gianni that he has a child'
- b. Marco *ha* *convinto* *Gianni a* *avere* *un figlio*.
 Marco₁ have-PRS.3SG convince-PST.PTCP Gianni₂ *a* PRO₁ have-INF a child
 'Marco has convinced Gianni to have a child'

In (3a), Mario causes Gianni to have a belief about some state of affairs *s*, whereas in (3b) the result is that Gianni has an intention to bring *s* about. In this paper, we look into the syntactic differences between these two infinitival structures, providing an account for their respective semantics.

2 Previous accounts on the belief/intention alternation

In recent literature, belief/intention alternation has been discussed in the broader context of mood alternation. As illustrated in (2), the belief versus intention readings with Greek *pitho* correlate with mood choice (Indicative vs. Subjunctive) introduced by different complementizers (*oti* vs. *na*, respectively). In the framework of Giannakidou & Mari, Indicative selection presupposes subjective veridicality, namely that the embedded clause is evaluated as true relative to the modal background introduced by the embedding verb. Subjunctive selection, by contrast, signals a suppositional attitude with epistemic uncertainty regarding the truth of the complement.

Turning to English *persuade*, Grano (2018) synthesizes two influential ideas: first, that modal quantification in attitude reports is contributed by the complement clause rather than

by the attitude verb itself (Bogal-Allbritten 2016, Kratzer 2006, 2013, Moulton 2009); second, that infinitives may carry built-in modal semantics (Bhatt 2006, Grano 2016). Leveraging these insights, Grano (2018) derives the alternation from the semantics of the complement: finite clauses introduce *information* modality, yielding a cause-to-believe reading, whereas (*for/to*) nonfinite clauses introduce *preference* modality, yielding a cause-to-intend reading.

- (4) a. $\llbracket \text{PRO to leave} \rrbracket = \lambda e. \forall w' \in \text{PREF}(e) : \exists e' [\text{Leave}(e') \ \& \ \text{Agent}(e, \text{PRO}) \text{ in } w']$
 b. $\llbracket \text{that it was raining} \rrbracket = \lambda e. \forall w' \in \text{INFO}(e) : \exists e' [\text{Rain}(e') \text{ in } w']$

On this account, both complements are eventuality descriptions, which allows *persuade* to maintain a single lexical entry in (5) targeting **rational attitudes** (a class including belief and intention, but not desire; see Grano 2015, 2018).

- (5) $\llbracket \text{persuade} \rrbracket^w = \lambda P_{vt}. \lambda x. \lambda y. \lambda e. \exists e'. \text{Persuade}(e) \wedge \text{Agent}(e, y, w) \wedge \text{Patient}(e, x, w) \wedge \text{CAUSE}(e, e') \wedge \text{RATIONAL-ATTITUDE}(e') \wedge \text{Experiencer}(x, e') \wedge P(e')$

According to (5), *persuade* takes as input an eventuality description and two individuals and returns an eventuality description: a persuasion event with agent y and patient x that causes an attitude eventuality e' with property P and experiencer x . Crucially, the attitude introduces the eventuality argument that serves as the modal anchor for the embedded quantification (Alonso-Ovalle, Menéndez-Benito & Rubinstein 2024, Hacquard 2010).

A limitation of this proposal, noted by Grano (2024), is that (4a) does not capture the satisfaction conditions of intention. As Searle (1983) observes, beliefs are satisfied by truth of content, but intentions require that the action is causally related to the intention itself. To address this, Grano (2024) propose that complements allowing intention readings abstract over an eventuality argument. In Romance, this correlates with mood choice, while in English it aligns with the finite versus non-finite distinction. The key refinement is **causal self-referentiality**: the state introduced by the attitude must causally contribute to the realization of the intended event described by the complement. This dependency is encoded in the semantics of *intend* (adapted in 6), where CAUSE^* connects the attitude state to the embedded event, and the complement supplies an abstracted eventuality argument for binding.

- (6) $\llbracket \text{intend} \rrbracket^w = \lambda P_{\langle e, \langle s, vt \rangle \rangle}. \lambda x_e. \lambda s_v. \exists s. \text{Intention}(s, w) \wedge \text{Experiencer}(s, x, w) \wedge \forall \langle w', y \rangle \in \text{CONTENT}(s) : \exists e. \text{CAUSE}^*(s, e, w') \wedge P(y)(w')(e)$

Indicative mood, by contrast, provides existential closure of the eventuality variable, which aligns with belief readings. However, this mapping does not straightforwardly extend to Italian, where the belief/intention alternation often appears with two infinitival types rather than a finite versus non-finite contrast: *di*-infinitives pattern with belief readings even though they do not combine with Indicative mood.

We therefore build on an alternative mentioned by Grano (2024) (following a suggestion by Menéndez-Benito): closure of the eventuality argument is effected by a dedicated head located structurally above Subjunctive but below Indicative. If independent evidence shows that *di*- versus *a*-infinitives differ in structural size, this mechanism extends naturally to the non-finite domain, deriving the observed semantic alternation from the interaction of complement size, eventuality abstraction, and the locus of modal quantification.

3 Evidence from structural size

In fact, the two infinitivals seem to exhibit crucial structural differences. To begin with, *di*-infinitives allow modal auxiliaries such as *potere* ('can') and *dovere* ('must') intervening between *di* and the infinitive, contrary to *a*-infinitives:

- (7) Marco ha convinto Gianni *di*
 Marco have-PRS.3SG convince-PST.PTCP Gianni *di*
 voler/poter/saper/dover studiare.
 want-INF/can-INF/being-able-INF/have-INF study-INF
 'Marco has convinced Gianni that he {wants to}/{can}/{knows how to}/{has to} study'
- (8) *Marco ha convinto Gianni *a*
 Marco have-PRS.3SG convince-PST.PTCP Gianni *a*
 voler/poter/saper/dover studiare.
 want-INF/can-INF/being-able-INF/have-INF study-INF
 'Marco has convinced Gianni to {wanting to}/{can}/{knowing how to}/{having to} study'

Since modal auxiliaries merge above the verbal domain (Cinque 2006, Grano 2015), their presence in *di*-infinitives but not in *a*-infinitives suggests that *di*-infinitives project a larger structure. This aligns with theoretical accounts linking structural size to propositionality (Lohninger & Wurmbrand 2020).

Moreover, we have seen that only *di*-infinitives can exhibit both subject and object control, whereas *a*-infinitive only allows object control (as shown in (3b)/(3a)). This follows if we assume that the former represent a standard case of control infinitives, with the possibility of projecting up to FinP — the projection that hosts the logophoric center (cf. Bianchi et al. 2003, Landau 2025, Rizzi 1997, a.o.) — and hence can mediate the control relationship between both the subject and the object. On the other hand, *a*-infinitives are expected to enforce object control if they lack this projection, given their reduced size: in such cases, the matrix object is simply the closest referent available for coindexation with the infinitival PRO. Momentarily leaving aside *di*-infinitives (as we assume constitute a trivial case of propositional infinitival complements), we will now focus on the syntactic and semantic properties of *a*-infinitives.

3.1 Zooming-in on *a*-infinitives

We begin by noticing that passive is available (9), as well as low aspectual restructuring verbs (10), which are merged at the edge of the vP domain (Cardinaletti & Shlonsky 2004, Cinque 2006, Grano 2015, Wurmbrand 2001).

- (9) La conduttrice ha convinto l'ospite ad essere
 the conductor have-PRS.3SG convince-PST.PTCP the guest TO be-INF
 intervistato.
 interviewed-PST.PRT.M
 'The conductor has convinced the guest to be interviewed.'

- (10) Marco ha convinto Gianni a cominciare a/continuare
 Marco have-PRS.3SG convince-PST.PTCP Gianni *a* begin-INF TO/continue-INF
 a/finire di lavorare.
 TO/finish-INF OF work-INF
 ‘Marco has convinced Gianni to start/continue/finish working.’

Interestingly, the infinitive in (9) necessarily implies an active (consenting) role in the interviewing event: Gianni does not passively enable it but rather allows it (it can be paraphrased as *Marco has convinced Gianni to let (someone) interview him*). This is consistent with the idea that, with *a*-infinitives, the attitude holder (Gianni in (9)) has an intention of bringing about the event described in the complement clause.

The reduced structure of these infinitives raises the question of how a future interpretation is licensed. In fact, root modality, which is located in the lower portion of the T layer, is associated with future orientation (Hacquard 2006). However, as we have seen, this position is not available in *a*-infinitives. By contrast, aspectual operators are typically merged even lower than root modals and may therefore remain accessible in these reduced structures. Interestingly, some aspectual operators have been argued to involve modal displacement: for instance, it has been proposed that the English progressive introduces ‘inertia worlds’ (Copley 2014, Dowty 1979, Portner 1998). A natural solution, then, is to posit the presence of an underlying future-oriented modal operator, similar to WOLL (Abusch 2004, Wurmbrand 2014). This operator would function as a factual modal (see Kratzer 2013), with a circumstantial modal base consisting of possibilities that share the relevant facts with the actual situation. This feature, incidentally, would also correctly predict that *convincere-a* reports would not be sensitive to *de dicto/de re* ambiguities. Consider the following context: *Marco has told Gianni that Luca is the new boss, and that he should talk with him. Gianni believes Marco, and he goes to talk to him. However, Luca is not the new boss, and Marco told him a lie.*

- (11) *Marco ha convinto Gianni a parlare con il nuovo capo
 Marco have-PRS.3SG convince-PST.PTCP Gianni TO speak-INF with the new boss
 ‘Marco convinced Gianni to speak with the new boss.’

The ungrammaticality of (11) arises because when *a* is present, the definite description *the new boss* must refer to the entity in accordance with the speaker’s circumstantial knowledge, despite Gianni’s mistaken beliefs about who the new boss is. Note that *di-infinitive*, meaning *Marco has convinced Gianni that he is talking with the new boss* is perfectly fine with a *de dicto* reading of the definite description (arising from Gianni’s beliefs), consistent with its assumed propositional nature. Another indirect argument in favor of a low, circumstantial modal akin to WOLL resides in the lack of restructuring effects, despite the reduced nature of these complements. According to Cinque (2004, 2006), Wurmbrand (2001), and Wurmbrand & Lohninger (2019), transparency effects such as clitic climbing are evidence for the lack of a complete functional structure above the infinitival verb. Consider (12):

- (12) L’operaio lo_1 prova a costruire t_1 .
 the-worker it_1 try-PRS.3SG TO build $_1$
 ‘The worker tries to build it.’

The object clitic *lo* is able to climb over the matrix verb due to the fact that both the restructuring verb *tries* and the lexical verb *build* are in the same domain, and the infinitive is not higher than a vP. The case of *a*-infinitives embedded under the attitude verbs addressed in this paper, however, is different in that it does not allow clitic climbing.

- (13) *Gianni lo_1 pensa a costruire t_1 domani.
 Gianni it_1 think-PRS.3SG TO build t_1 tomorrow
 ‘Gianni thinks about building it tomorrow.’

We can account for this restriction if (12) involves an underlying modal element that provides the future orientation of the complement clause, essentially constituting two different (temporal) domains. This also explains why it is possible to have two distinct temporal adverbials in the matrix and complement clauses (14) – a pattern not found in standard restructuring configurations, where the infinitival complement lacks independent temporal specification and is fully dependent on the temporal properties of the restructuring verb (15).

- (14) Ieri Gianni mi ha convinto a comprare una macchina
 yesterday Gianni me.CL have-PRS.3SG convince-PST.PTCP TO buy-INF a car
 nuova il mese prossimo.
 new the month next
 ‘Yesterday Gianni convinced me to buy a new car next month.’

- (15) *Ieri Gianni ha provato a riparare la macchina il mese
 Yesterday Gianni have-PRS.3SG try-PST.PTCP TO repair the car the month
 prossimo.
 next
 ‘Yesterday Gianni has tried to repair the car next month.’

Our structural hypothesis is further supported by the fact that these *a*-infinitives can host negation (16), indicating that they must project a (reduced) TP. This makes them comparable to the *situation* complements described Wurmbrand & Lohninger (2019) and Lohninger & Wurmbrand (2020), separating them from restructuring infinitives which would be in fact *event* complements.

- (16) Gianni mi ha convinto a non salutare più il capo.
 Gianni me.CL have-PRS.3SG convince-PST.PTCP TO not greet-INF anymore the boss
 ‘Gianni has convinced me not to say hello to the new boss anymore.’

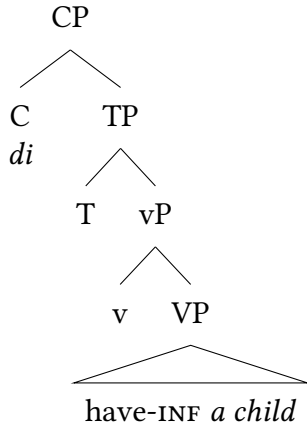
Finally, only *di*-infinitives can be assessed for truth (17), a behaviour expected if they are propositional complements. By contrast, the fact that *a*-infinitives cannot be assessed for truth suggests that they are not structurally large enough to constitute propositions (cf. Wurmbrand & Lohninger 2019).

- (17) a. Marco ha convinto Gianni di avere un figlio, ma non
 Marco have-PRS.3SG convince-PST.PTCP Gianni DI have-INF a child, but not
 è vero.
 be-PRS.3SG true
 ‘Marco has convinced Gianni that he has a child, but it is not true.’

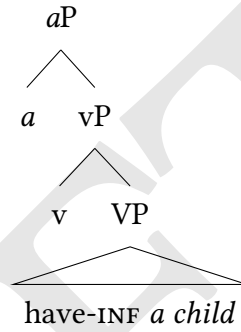
- b. Marco ha convinto Gianni ad avere un figlio, # ma non
 Marco have-PRS.3SG convince-PST.PTCP Gianni A have-INF a child, but not
 è vero.
 be-PRS.3SG true
 ‘Marco has convinced Gianni to have a child, but it is not true.’

Our exploration into the structural size of the two infinitival complements leads us to the following (sketched) representations.

- (18) *di avere un figlio*



- (19) *ad avere un figlio*



4 Semantic analysis

The difference in size between the two infinitival structures has important semantic consequences (Lohninger & Wurmbrand 2020). We propose that *a* selects an event-denoting infinitive whose eventuality argument remains unsaturated (Grano 2024), while *di* selects a larger, propositional structure in which the eventuality argument is saturated via a CLOSURE operator. This distinction is reflected in the denotations in (20).

- (20) a. $\llbracket [\text{vP} \text{PRO vincere}] \rrbracket = \lambda x. \lambda w. \lambda e. \text{WIN}(e, w) \ \& \ \text{AGENT}(x, e, w)$
 b. $\llbracket [\text{TP} \text{CLOSURE PRO vincere}] \rrbracket = \lambda x. \lambda w. \exists e'. \text{WIN}(e', w) \ \& \ \text{AGENT}(x, e', w)$

Following Grano (2024), intention encoding requires eventuality abstraction, so that the embedded event is causally linked to the matrix attitude. In the Italian non-finite domain, we argue that this causal link is introduced by *a*, yielding a ‘cause-to-intend’ reading, while *di* selects a proposition with an abstracted eventuality argument. Assuming, with Kratzer (2006) and Moulton (2009), that modal quantification arises inside the embedded clause, we posit a single meaning for *convincere* as “cause to have a rational attitude”, with finer meaning differences stemming from the type of complement. This unified lexical entry will be slightly revised in §5.

We adopt a single denotation for *convincere*, building on Grano’s (2017, 2019) proposal for English *persuade*. On this view, *convincere* introduces a causal relation between an event of convincing and a resulting mental state classified as a rational attitude. Rational attitudes

in this sense include beliefs and intentions, but crucially not desires, which are excluded because they lack the rational evaluability necessary for the argumentative structure targeted by *convincere*. The denotation we adopt is given in (21).

$$(21) \quad \llbracket \text{convincere} \rrbracket^w = \lambda P_{vt}. \lambda x. \lambda y. \lambda e. \exists e'. \text{Convince}(e) \wedge \text{Agent}(e, y, w) \wedge \text{Patient}(e, x, w) \wedge \text{CAUSE}(e, e') \wedge \text{RATIONAL-ATTITUDE}(e') \wedge \text{Experiencer}(x, e') \wedge P(e')$$

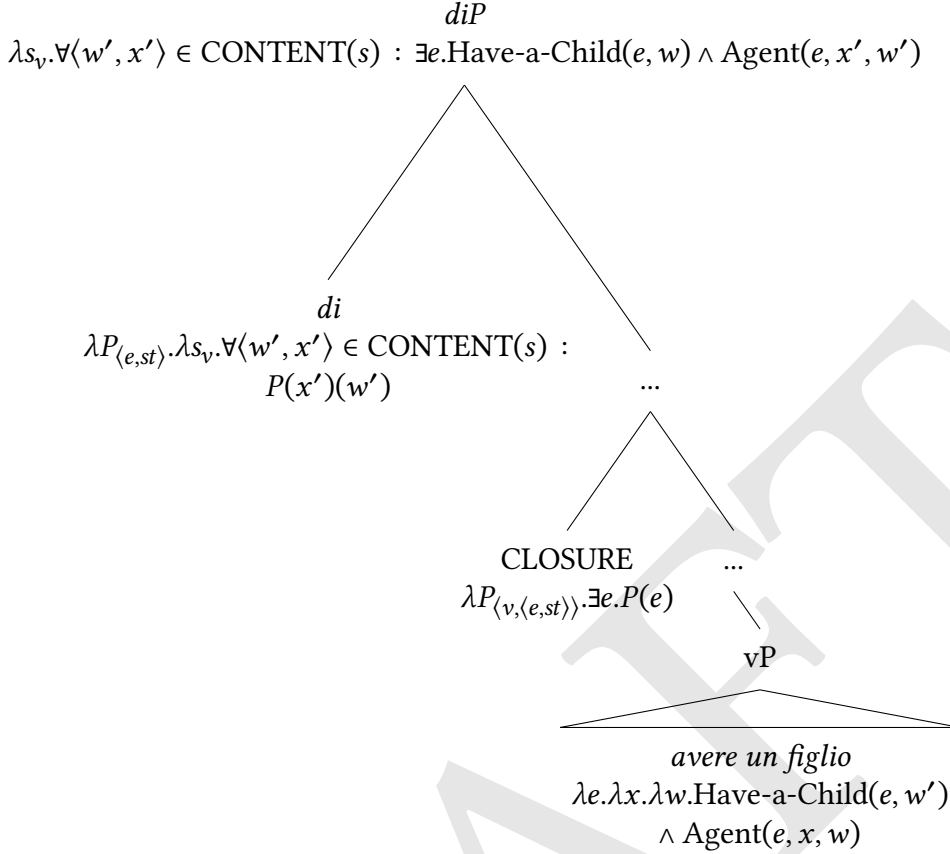
This lexical entry provides the core meaning of *convincere* independently of the type of infinitival complement. The differences between *a*- and *di*-infinitives emerge from the size of the complement and the way the eventuality argument is handled.

The bare vP infinitive in Italian denotes a proposition abstracting over individual, world, and eventuality arguments (Chierchia 1989, Pearson 2013). When selected by *a*, this minimal structure receives additional functional structure introducing a causal relation. Specifically, *a* introduces a CAUSE* relation that binds the eventuality argument, thereby linking the intended event of the embedded clause to the attitude holder. This mechanism accounts for the causal self-referentiality of intention, namely, the intuition that the attitudinal state must causally contribute to the realization of the intended event. Additionally, *a* also introduces quantification over possible worlds, following Kratzer's (2006) and Moulton's (2009) insight that the attitude is located inside the complement clause, rather than in the attitude verb. The tree in (22) illustrates the composition for a case with an *a*-infinitive.

$$(22) \quad \begin{array}{c} aP \\ \lambda s_v. \forall \langle w', x' \rangle \in \text{COMPATIBLE}(s) : \\ \exists e. \text{CAUSE}^*(s, e, w') \wedge \text{Have-a-Child}(e, w) \wedge \text{Agent}(e, x', w') \\ \swarrow \quad \searrow \\ a \quad \quad \quad vP \\ \lambda P_{\langle v, \langle e, st \rangle \rangle}. \lambda s_v. \forall \langle w', x' \rangle \in \text{COMPATIBLE}(s) : \\ \exists e. \text{CAUSE}^*(s, e, w') \wedge P(e)(x')(w') \quad \swarrow \quad \searrow \\ \quad \quad \quad \text{avere un figlio} \\ \quad \quad \quad \lambda e. \lambda x. \lambda w. \text{Have-a-Child}(e, w) \\ \quad \quad \quad \wedge \text{Agent}(e, x, w) \end{array}$$

By contrast, *di* selects a larger clausal structure capable of hosting additional functional projections. The crucial difference is that the eventuality argument of the embedded clause is bound by an intervening CLOSURE operator. This operator existentially closes the event variable, yielding a propositional structure that abstracts only over the individual and world arguments. As a result, the complement of *di* denotes a saturated proposition, rather than a minimal eventive property. This is shown in (23).

(23)



In sum, both *a*- and *di*-infinitives yield properties of states, which can combine with the lexical meaning of *convincere*. The difference lies in whether the eventuality argument is left open and linked causally via *a*, or instead bound internally by a CLOSURE operator under *di*. This compositional distinction explains why both complements are compatible with a single lexical entry for *convincere*, while still producing the observed interpretive contrasts in the Italian infinitival domain.

5 Conclusions

This paper has examined the belief/intention alternation in Italian embedded infinitives, focusing on the syntactic and semantic distinctions between the two types of infinitival complements selected by verbs like *convincere* ('convince'). We have shown that the alternation, which is commonly attributed to finiteness and mood distinctions in other languages, is instead mapped onto the structural size of the complement. The analysis presented here suggests that *di*-infinitives are associated with larger structures, enabling the embedded clause to host additional functional projections that allow for propositional interpretation and eventuality closure. In contrast, *a*-infinitives are structurally smaller, linking more directly to the causal intention of the matrix verb and implicating a more minimalistic event structure. These differences align with the belief versus intention readings observed in the two constructions, where *di*-infinitives convey a belief-like interpretation, while *a*-infinitives give rise to an intention interpretation that involves causal self-referentiality.

Abbreviations

1 = first person, 2 = second person, 3 = third person, acc = accusative, cl = clitic pronoun, gen = genitive, inf = infinitive, ind = indicative, m = masculine, pl = plural, prs = present, pst = past, ptcp = participle, prt = participle, sg = singular, sbjv = subjunctive.

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